

## Origami Fencing from Waste Cans

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### Background

#### Solid Waste disposal in Sub-Saharan Africa

Proper management and disposal of solid waste is crucial, as open dumping and burning have enormous negative impacts both on public health and on the environment (Adedara et al., 2023; Ferronato & Torretta, 2019). The annual waste generation in sub-Saharan Africa has been increasing and is expected to keep doing so as populations and economies grow, while solid waste collection and coverage rates are still far below satisfactory at around 65% (Ferronato & Torretta, 2019).

#### Aluminium Recycling in Malawi

Aluminium and other scrap metal is collected in Malawi by waste pickers but is mostly exported to other countries rather than being recycled in Malawi (Kasinja & Tilley, 2018). It is also frequently turned into artisanal cookware, which is a health risk due to metal leaching into the food when using the recycled cookware (Street et al., 2020).

#### Resource shortages in Malawi

Malawi suffers from resource shortages, leading to delays in construction projects and making construction materials essentially unavailable for many (Moses & Wynand, 2013). Theft of valuable resources such as aluminium cables only worsen the situation (Beston, 2025). Farmers that require fencing to protect their produce from grazing goats cannot afford to build a fence from conventional materials.

#### Requirements for goat suitable fencing

The practise of letting goats roam free leads to the problem of farmers crops being consumed and destroyed by the goats (Rumosa Gwaze et al., 2008). To protect the crops from the goat herds, suitable fencing must be used. The fencing must not present a danger to the goats, e.g. it mustn't have sharp edges that injure the animals or have loose strings in which they can entangle and hurt themselves (Paige, 2008). At the same time, the fencing needs to be high enough to stop the goats from jumping it, and reach to the ground so the animals cannot slip under it (Lund & May, 2003).

#### Aluminium can upcycling

Various methods to upcycle aluminium beverage cans already exist and have been documented online. Project Kamp, a community research project for more sustainable living, has built a roof from shingles made up of aluminium cans that were cut up and flattened (Cheung & Hakkens, 2023). Keving Cheung designed a can slicer, intended to quickly and cleanly cut off the top and bottom parts of aluminium cans, leaving a sheet metal cylinder (Cheung, n.d.). Other creators turned used beverage cans into low voltage electric wiring (CircuitsReuse, n.d.) or even jewellery.

## Problem statement and goals

Aluminium beverage cans in Malawi are being collected, but do not get recycled. Farmers need fencing to protect their crops from roaming goats, but can't afford them. The goal of this project is to design a method to upcycle empty aluminium beverage cans into fencing. The fencing design has to be robust enough to hold back goats without posing an injury risk to the animals. A prototype machine for the cutting and folding of the cans is to be developed. The machine must be simple to manufacture in low resource environments and easy to operate and maintain.

## Research Questions

1. How can aluminium beverage cans be turned into fencing suitable for managing goats?
2. What is the tensile strength that can be achieved with a fence made of aluminium wiring?
3. How can a machine facilitate and speed up the process of fence manufacturing?

## Methods

1. Requirements for goat fencing will be researched and commercially available fencing designs will be studied. Aluminium beverage cans will be cut into strips or sheets. A method for interlocking the strips to form a wire will be determined experimentally. Inspiration from origami techniques could be used to find a satisfactory method. The aluminium wiring could then be woven into mesh to increase the fence's strength. Metrics to categorise the resulting fence designs could include the tensile strength, how much of an individual can actually be used (or how much of it ends up as scrap), how many steps are required in the manufacturing process or how many cans are required per meter of fence.
2. The tensile strength of the resulting fence designs will be measured. Additionally, the durability to different kinds of stresses could be tested, such as simulating goats chewing on the wiring or ramming it. This may be a crucial weakness, as the thin aluminium from the cans is very prone to fatigue fractures after bending just a few times.
3. For the resulting fencing design, the relevant manufacturing steps will be analysed. A machine will be designed to aid in carrying out the manufacturing steps. A prototype of the machine will be built and tested. The machine will be designed to be easy to manufacture from locally available resources. It will be portable and powered by hand (e.g. via a hand crank) to allow it to be used anywhere. The prototype of the machine could then be further improved and implemented in Blantyre, Malawi, as part of a master's thesis. The prototype will presumably be built using 3D printed and / or laser cut parts, to allow for rapid prototyping.

## Planning

### Gantt chart

I am doing this semester project part time, from the 02.03. to the 29.05.

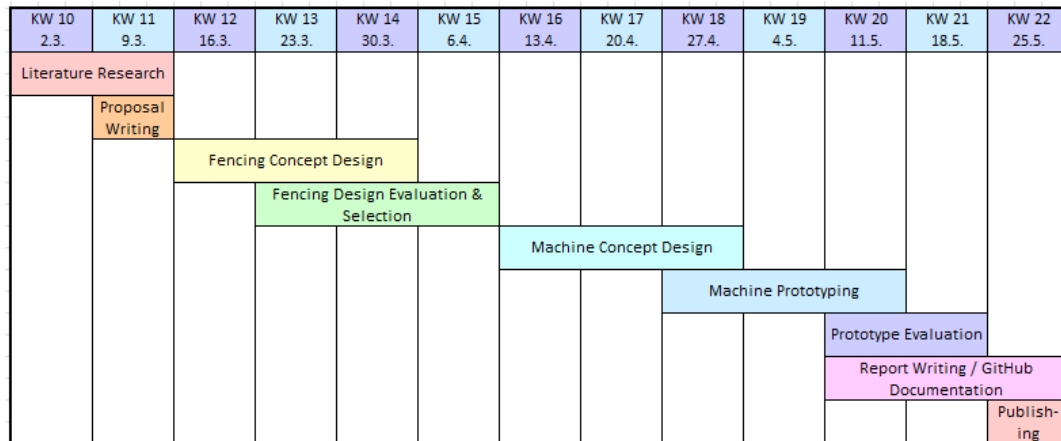


Figure 1. Gantt chart for research work and thesis writing.

After some initial research I will start to develop and build some fence concepts from cans by hand. The robustness and strength of the fence designs will then be tested and the results used to improve the designs. It may turn out that aluminium breaks too quickly from fatigue, which could be a potential showstopper. Once the best fence design is determined (one that is not too complex to build in as few as possible steps, yet robust enough), a concept for how to mechanise the production will be developed. I plan to start prototyping early and quickly using 3D printed parts. The last two to three weeks I will focus on evaluating the prototype and reporting my results. If the results are promising I would like to then continue with the project for my master's thesis.

## Data Management Plan (DMP)

### Data collection

I am not planning to collect survey data. The only data I will be dealing with is the fencing designs and machine parts that will be developed. Some variables may be used to characterize and evaluate the fencing designs such as:

- Tensile strength
- Material cost (Cans / meter)
- Manufacturing complexity
- Efficiency in can usage (how much of a can is actually used for the fencing)

| dataset                      | data type                                                                                                                                                                                                                                               | data format              | data documentation                         |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------|
| <i>Fence designs</i>         | <i>Concepts of fencing designs</i>                                                                                                                                                                                                                      | <i>Drawings / Images</i> | <i>Stored in a document on GitHub repo</i> |
| <i>Fence Characteristics</i> | <i>Variables describing the performance of the fence designs:</i> <ul style="list-style-type: none"> <li><i>Tensile strength</i></li> <li><i>Material Cost</i></li> <li><i>Manufacturing Complexity</i></li> <li><i>Can usage Efficiency</i></li> </ul> | Excel table              | <i>Stored in a document on GitHub repo</i> |
| <i>CAD Drawing</i>           | <i>Model of the Machine</i>                                                                                                                                                                                                                             | .stl files               | <i>Stored on GitHub repo</i>               |

### Data storage

The fence designs and their characteristic variables will be stored in an excel sheet, where they can easily be compared to each other. The CAD Drawings of the machine prototype will be stored in a GitHub repository that will be used to document the entire project.

### Budget

I can only give a rough estimate for the costs of the prototype, but I expect to remain under 100 CHF for the total material costs. The main expenses will be the knives for the cutting of the aluminium, plywood to build a solid base and some screws and nuts to fit the parts together.

| No           | expense               | amount | unit price [CHF] | total price [CHF] |
|--------------|-----------------------|--------|------------------|-------------------|
| 1            | 3D Printing Filament  | 1 kg   | 20               | 20                |
| 2            | Knives / Razor Blades | 5      | 5                | 25                |
| 3            | Screws etc.           | 20     | 2                | 40                |
| 4            | Plywood               | 1      | 15               | 15                |
| <b>Total</b> |                       |        |                  | <b>100</b>        |

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